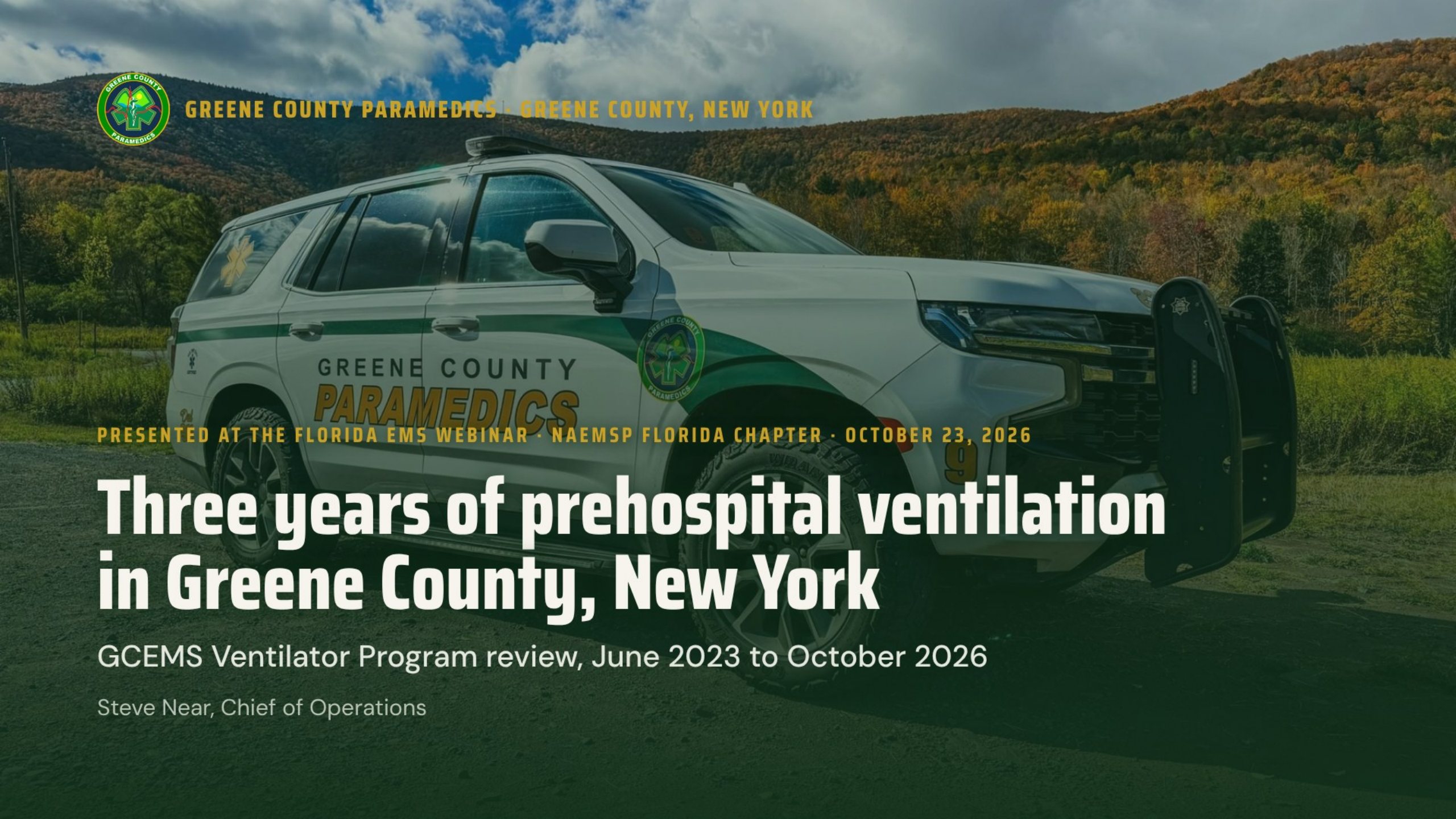


GREENE COUNTY PARAMEDICS · GREENE COUNTY, NEW YORK

PRESENTED AT THE FLORIDA EMS WEBINAR · NAEMSP FLORIDA CHAPTER · OCTOBER 23, 2026

Three years of prehospital ventilation in Greene County, New York

GCEMS Ventilator Program review, June 2023 to October 2026

Steve Near, Chief of Operations

ABOUT THE PRESENTER

Steve Near

Chief of Operations, Greene County Paramedics, since 2022. Paramedic since 2014. I staff Medic 1, the chief's vehicle. Like every other car, it carries a ventilator, and I still use it on 911 calls.

> **Before GCEMS:** ten years at Greenport Rescue Squad across the Hudson, a 7,000 to 8,000 call a year agency. 911 paramedic, duty supervisor from 2015, and critical care interfacility transports including ventilated patients.
>
> **At GCEMS since 2022:** built the prehospital ventilator program; kept an existing RSI program current with multiple mandatory trainings a year; Sapphire infusion pumps on every car by the end of 2026.
>
> **Regional:** REMSCO committee member for the REMO Hudson-Mohawk Valley region, and a member of the region's current prehospital blood committee.

REMO Harriet C. Webster Award for Leadership, 2024. During the program's first years GCEMS received REMO's Agency of the Year (2023) and Excellence in Quality and Innovation (2024).





Right: with agency president Mark Evans after the REMO awards.

WHO WE ARE

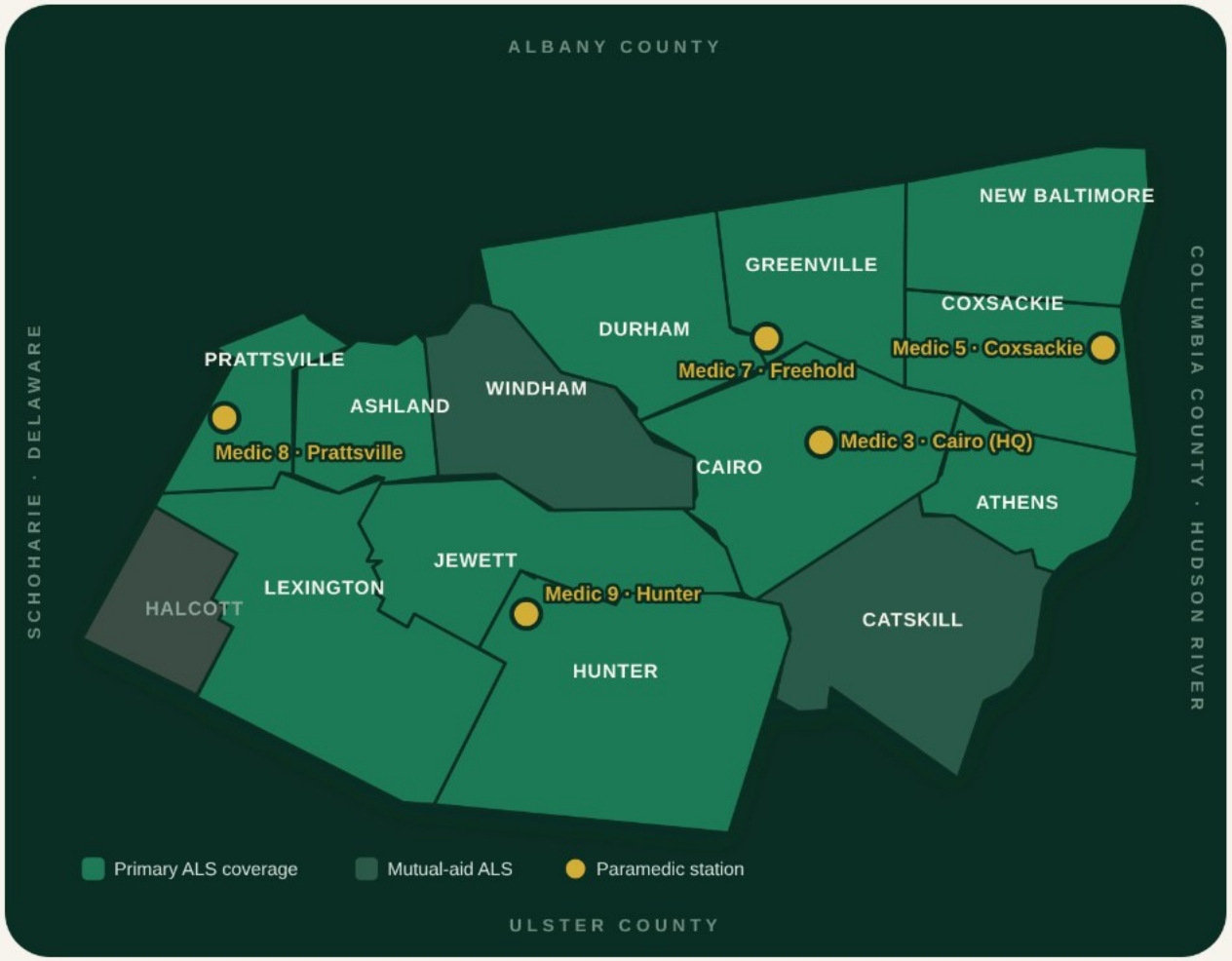
Greene County Paramedics

Greene County Paramedics is an ALS first-response fly-car agency. We do not transport. We respond in Chevy Tahoes from five stations, meet the local ambulance on scene, and ride with the patient to the hospital. Our closest hospital is Columbia Memorial in Hudson; the nearest trauma center, cath lab and tertiary care are at Albany Med, which from the mountaintop could be more than 80 minutes by ground, and a good part of our winter is not flyable.

658
square miles, northern Catskills

4,125
dispatches in 2025, 911 only

40
paramedics on staff



Five stations, six units. Medic 1, the chief's vehicle, is based at Cairo headquarters. Town boundaries simplified.

WHY WE STARTED THIS

Why should care have to change hands for the patient to receive the care they need?

As paramedics we could arrive at someone's house, resuscitate or treat them, and bring them to the hospital using a BVM or CPAP. Normal practice.

Later that same shift I could be sent to move that same patient to another facility on a ventilator, making every clinical decision and every setting change myself for a 45-minute ride.

If you traveled to any surrounding 911 EMS Agency, you no longer had that autonomy or capacity.

That never sat right with me. Nothing in the protocols said we couldn't. Nobody had thought a ground 911 agency would, so nobody had written it down. So we did it.

MORNING · 911 CALL

Resuscitate or treat. Transport with a BVM or CPAP. Normal practice.

SAME SHIFT · INTERFACILITY TRANSFER

Same patient, on a ventilator. Every setting change mine for 45 minutes. Also normal practice.

> **"Just because a paramedic can doesn't mean a paramedic should."**
>
> What we heard in 2022. The rest of this talk is the answer.

HOW WE BUILT IT

Program timeline, 2023 to 2025

APRIL 2023

CMH donates LTV 1200s

Retired transport vents, free. All 40 providers trained.

JUNE 2023

LTVs go in service

Three cars with the widest coverage. Invasive only, written for the RSI patient.

OCTOBER 2023

First survivor goes home

Post-ROSC on the LTV to the cath lab. Arrest was never the plan. It changed the program.

JUNE 6, 2024

Hamilton T1 ×3 go live

Medics 3, 5 and 9. LTVs move to the other cars. First non-invasive capability (NIV-ST).

JANUARY 1, 2025

Full fleet on T1

Six ventilators, one in every car. LTVs retired. Every provider retrained before go-live.











Left to right: the donation from Columbia Memorial Health · the LTV kit · the LTV in its carrier · deployment day, June 2024 · the first Hamilton T1s. Green points are the LTV era; red are the T1.

YEAR ONE · 2023 TO 2024

Could a 911 paramedic run a ventilator for an hour?

> **A flight crew's first job on board was to get the patient onto their vent and dial the support back into reality. They were chasing their tail from the second they landed, because the patient had been bagged the whole way there. Maybe well. Probably not.**
>
> The plan was to do that step ourselves: RSI the patient, put them on our ventilator, and hand LifeNet or the ER a patient who is already dialed in.

Before the first LTV went on a car, every provider sat through multiple in-person classes: vent dynamics, pressure versus volume, SIMV versus assist control, troubleshooting alarms. In June 2023 three units went on the cars that reach the most of the county. Invasive only, written for the RSI patient.

- First use, two days after go-live: a polypharmacy overdose needing RSI, in territory we don't primarily cover. We were requested over the air for RSI and vent capability
- Twelve patients in year one, all of them intubated. The LTV couldn't do anything else
- Four months in, a crew used it on a patient we had never planned for. Next slide





The LTV kit, 2023: a recycled blue EMS bag with the ventilator, external battery, circuits, oxygen tubing and the alarm reference.

WHAT THE LTV TAUGHT US

The crews could do it, and the vent was good for more than RSI. The gap was the patient in respiratory distress who was still awake and still breathing on their own. We had nothing for them. That's where the T1 comes in.



OCTOBER 2023 · FOUR MONTHS IN · LTV 1200

The first one home.

A woman in her early fifties collapsed at home with no warning. Ventricular fibrillation when the crew arrived. Resuscitated on scene, intubated, and put on the LTV. The helicopter could not fly that night, so she went by ground, managed as a critical care patient for the whole ride to Albany Medical Center and the cath lab.

V-fib

on arrival. ROSC on scene.

Grounded

Weather kept the aircraft down. Ground, on the LTV, the whole way.

Home

Discharged neurologically intact.

We had written the program for the RSI patient. Cardiac arrest wasn't in the plan. A crew put her on the vent anyway, and she turned it into a ventilation program.

"We watched our survival rate climb at the end of 2023: four of nine arrests survived to discharge to close out the year. In 2024 we were three for five with ROSC patients ventilated from the field to the hospital, and all three were deficit-free."

From a note to another EMS agency, 2024.

Patient story shared without identifying details.

THE EQUIPMENT

Why the Hamilton T1

All twelve intubated patients in the LTV year were managed without complication. The crews had demonstrated they could run a ventilator, and the question changed from whether we could to how to make the program more robust. The patients we still couldn't help were in respiratory distress but not yet in respiratory failure: awake, fatiguing, and candidates for non-invasive support. The T1 does invasive and non-invasive ventilation on one device: NIV-ST, what most people call BiPAP, and the full invasive mode set. If a patient fails NIV we change the interface, not the machine.





MODES IN GCEMS USE

- **NIV-ST** — NIV · what most call BiPAP
- **ASV** — default invasive
- **APV-CMV** — volume
- **PCV+** — pressure
- **CPR mode** — during resuscitation

> **When we start NIV-ST:** at the same point the Collaborative Protocols call for CPAP, or earlier if the vitals and presentation say the patient is headed there. Same trigger, better support.

Reference cards on every ventilator: ideal body weight, alarms, starting settings. Aligned to the New York State Collaborative Protocols.

TRAINING AND OVERSIGHT

Everyone is trained. Every chart is read.

40 / 40
providers trained before each of three go-lives

2 / month
ventilator workshops, four providers at a time, on a running vent

Every chart
ventilator and NIV charts read by three people, as close to the call as possible

- Initial sessions taught by Hamilton Medical and LifeNet of New York alongside our ALS Coordinator; a required number of workshop sessions per provider each year
- Workshops are scenario-based: setup, alarms, troubleshooting, mode changes on a running ventilator
- RSI, a separate program under medical director credentialing: two mandatory sessions a year, a protocol update and a practical skills exam
- Chart review by the Chief of Operations, Deputy Chief and ALS Coordinator, with a daily check for any use not already reported up. Significant patients go to Dr. Craig Stanger, our medical director
- Findings go back to the provider and into the workshop scenarios. Reference cards ride on every ventilator







> **Forty trained. Every chart read. This is what three years of that looks like.**

THREE YEARS IN · JUNE 2023 TO OCTOBER 2026

A lot of agencies have put ventilators on trucks. Not many have published what happened next. This is everything we have, including the parts we're still working on.

6/6

units carrying a Hamilton T1

147

patients ventilated in the field

79

patients supported with NIV (NIV-ST) since June 2024

0

intubations for respiratory failure since April 2025

WHAT THE DATA SHOWS

Ventilator use by period

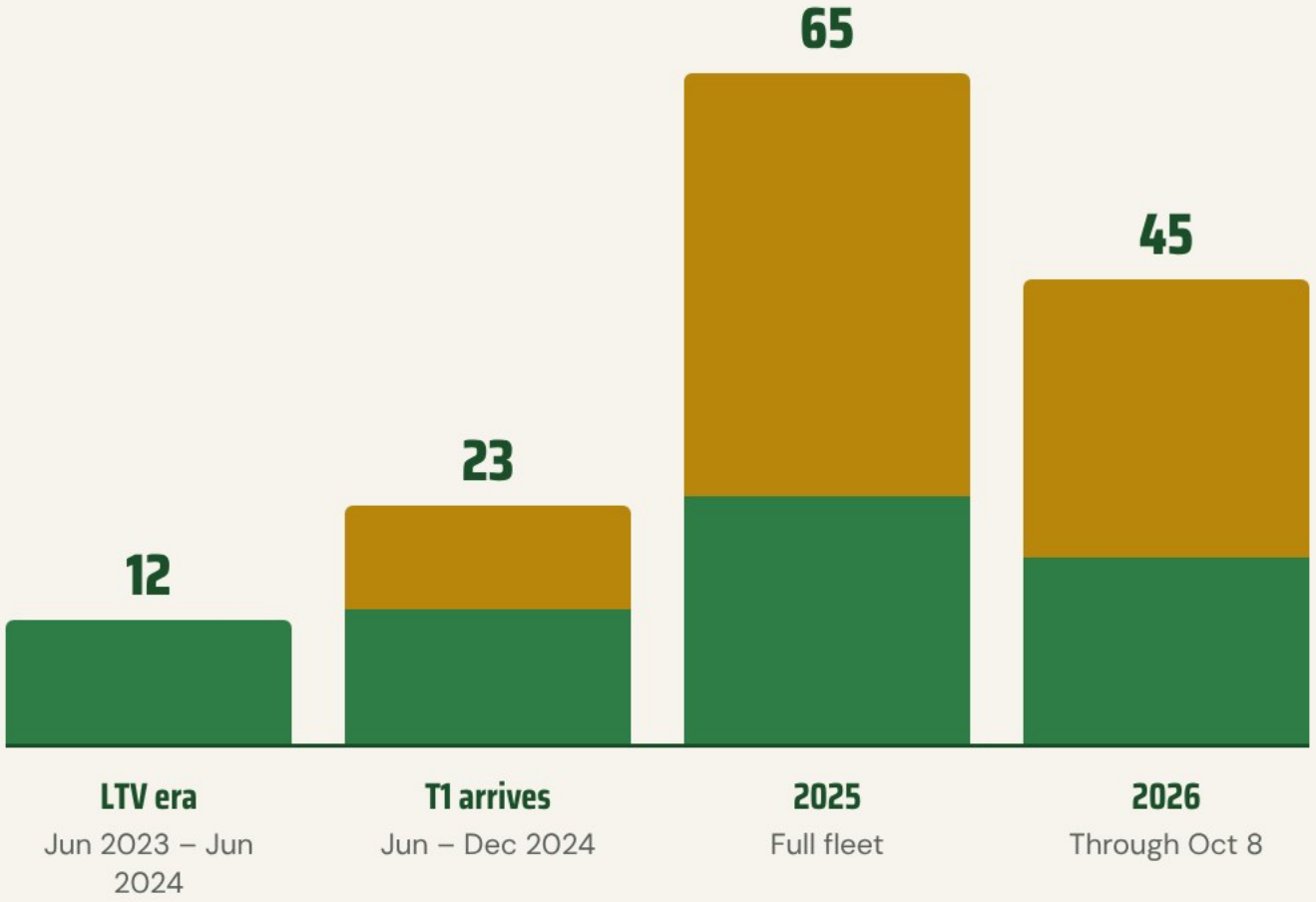
NIV opened the ventilator up to patients who, before, either kept struggling or got sedated and intubated.

- NIV-ST (non-invasive, BiPAP)
- Invasive (intubated)

Share of ventilator patients on NIV: 0% in the LTV era, 43% by late 2024, 63% in 2025, 60% in 2026.

ESO records, 911 responses only, June 2023 to October 8, 2026. Any charted ventilator or NIV treatment, one count per incident; CPAP alone not counted. The published review's manual count is 147 ventilated and 79 on NIV; an ESO query returns 145 and 78, the difference being cases charted under oxygen delivery. These slides use the published NIV count of 79.

Ventilator patients by period

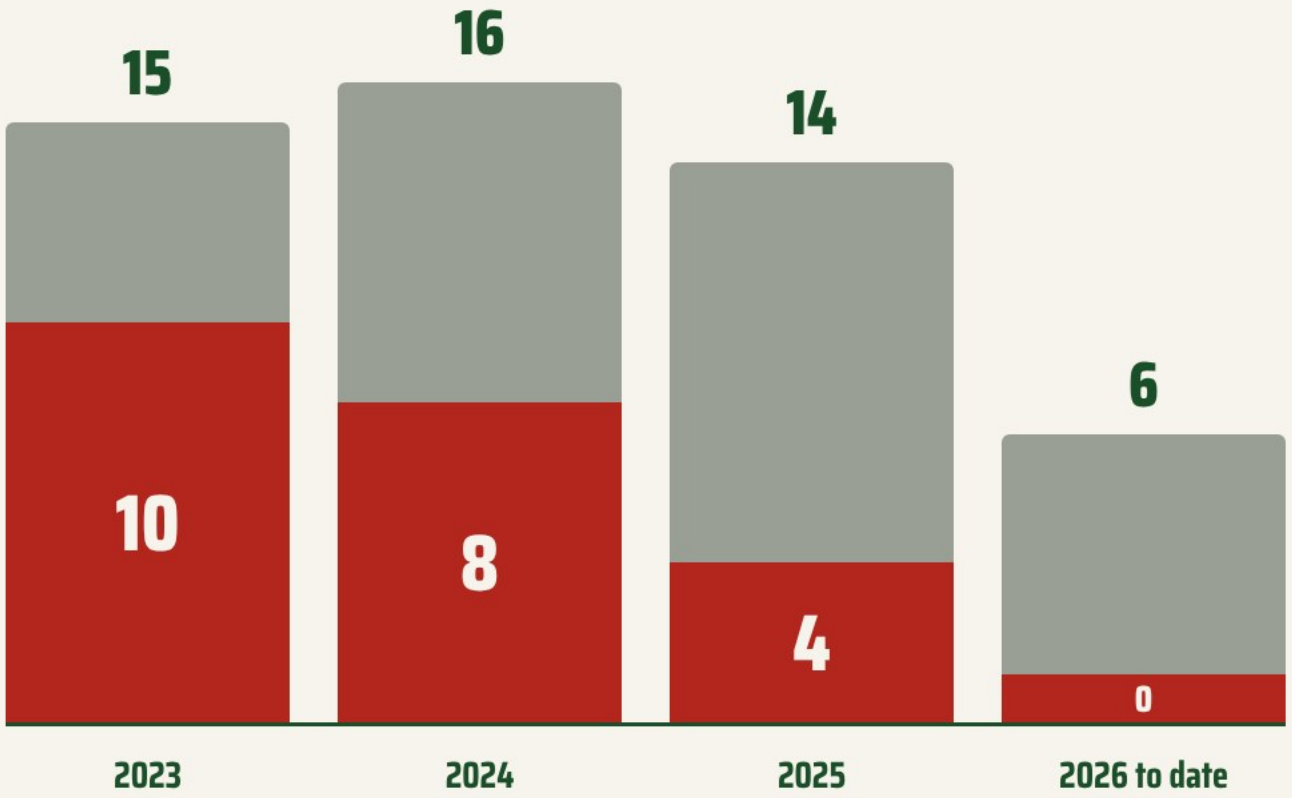


IF YOU ASK US **"Why NIV-ST rather than CPAP?"**

For CHF, CPAP is usually enough. For a tiring COPD or asthma patient, the inspiratory support is what keeps them off the tube, and on the T1 it is the same device if they fail.

Rapid sequence intubations by year

Total RSIs, with the respiratory-failure share in red



all 4 before April

RESPIRATORY INTUBATIONS

10 - 8 - 4 - 0

We still intubate when a patient needs it: trauma, cardiac arrest, altered mental status. What changed is respiratory failure. Since April 2025, every respiratory-failure patient who could go on NIV did, and none have needed a tube in the field. They aren't being held in respiratory failure until the hospital takes over. They're getting better on the way in, and the next slide shows by how much.

RSI counted as paralytic administration in the agency record. Respiratory failure classified by chart review.

IF YOU ASK US **"So you stopped intubating?"**

No. Trauma, arrest and altered mental status still get a tube; six RSIs this year. What changed is respiratory failure. RSI is counted as paralytic given; respiratory failure by chart review.

DID THEY GET BETTER?

93% of NIV patients improved before arrival.

Two separate measures: what the crew charted at handoff, and what the monitor recorded from first contact to arrival.

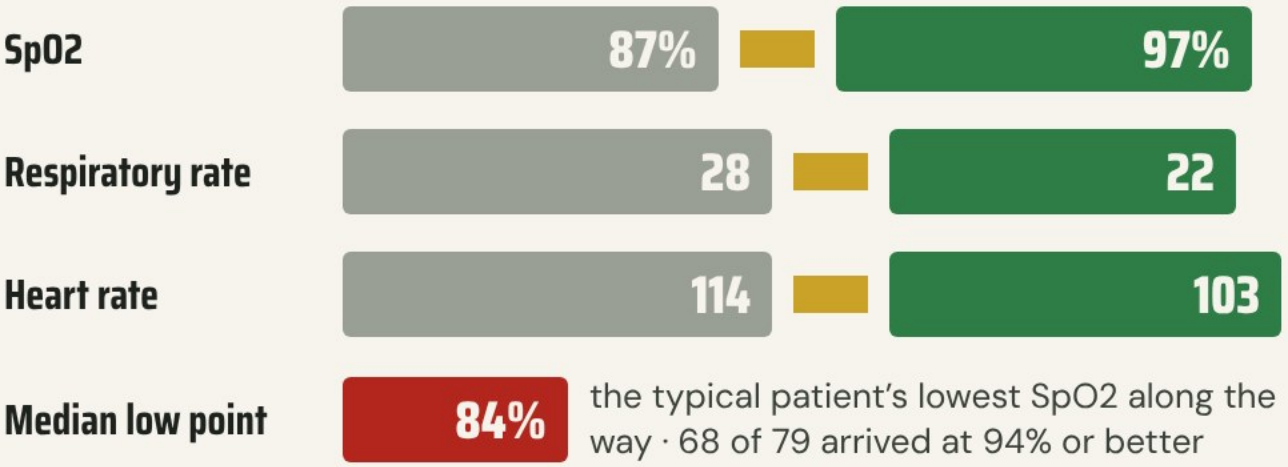
74 of 79

NIV patients charted "Improved" at destination; 4 unchanged, 1 could not tolerate the mask.

NIV patients: median vital signs, first charted vs. at handoff

Median age 70. Total range 13 to 94

Average of 23 minutes from patient contact to NIV,



IF YOU ASK US **"How do you know they improved and not just that you charted it?"**

Two separate sources. Crews charted 74 of 79 as improved, and the monitor data shows the same trend on its own. Source: ESO vitals, first and last values gathered, on the 79 patients who had them.

CASE · 2024 · EARLY NIV

Asthma, intubated twice before

A patient with severe allergic reactions and asthma, intubated twice before for the same thing, found barely conscious and struggling to breathe. Treated with NIV-ST and medication, improved during transport and arrived awake.

The mask came off about an hour after handoff and the patient was discharged that evening without being intubated. One of the first patients on the T1's non-invasive mode.



Right: the full T1 setup during a real NIV transport.

WHAT MANAGING THE PATIENT LOOKS LIKE

Two charts, two patients, as recorded

NIV-ST, TITRATED OVER THIRTY MINUTES

74-year-old, acute pulmonary edema, Athens to Columbia Memorial, March 2025. Four charted setting changes.

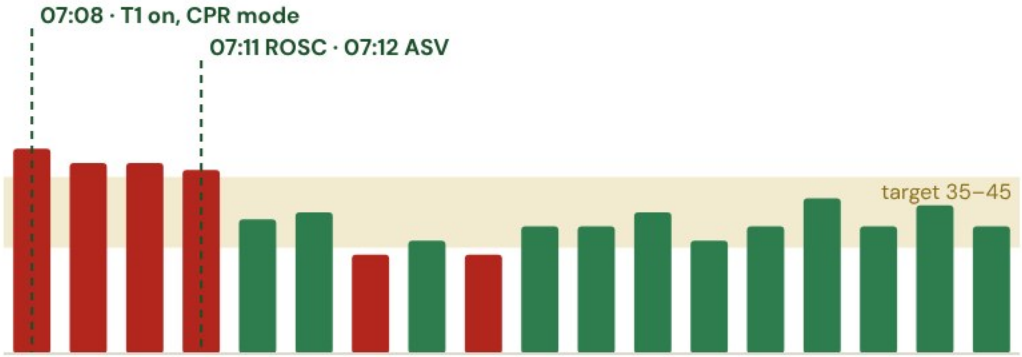
20:17	20:25	20:28	20:35	20:40	20:48
First vitals SpO2 72%, RR 28	**NIV-ST 10 / 5** FiO2 100%	**12 / 5** FiO2 100%	**12 / 5** FiO2 75%	**12 / 5** FiO2 50%	**Handoff** CMH

SpO2	Resp rate	Heart rate
72% → 97%	**28 → 19**	**134 → 76**

Started at 10/5, stepped to 12/5 once the mask was tolerated, then FiO2 weaned from 100% to 50% as saturation held. Charted "Improved" at destination.

POST-ROSC, ETCO2 HELD TO TARGET

79-year-old cardiac arrest, Greenville to Albany Medical Center, November 2025. ROSC at 07:11, cardiac alert, 35 minutes of transport.

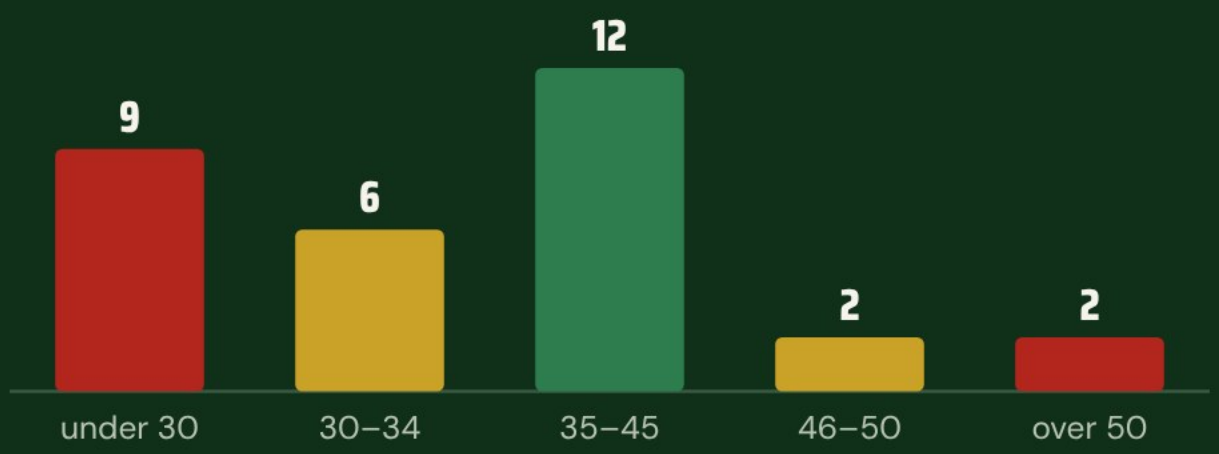


18 EtCO2 values in 55 minutes. T1 on in CPR mode at 07:08, ROSC 07:11, ASV from 07:12. Green bars are within 35–45 mmHg; red are outside. Charted "Improved" at destination.

INVASIVE VENTILATION · THE NUMBER WE ARE STILL WORKING ON

9 of 31 patients arrived with an EtCO2 under 30. We know why.

Arrival EtCO2, intubated patients not in cardiac arrest (31 of 32 had a reading at the door)



Median EtCO2 moved from 29 at the first reading to 35 at the hospital door, so crews are correcting on the way in. The nine under 30 we read as over-ventilation. Post-intubation EtCO2 and oxygen targets are our 2026 QA measure.

67 of 68
intubated patients had EtCO2 recorded
36 cardiac arrests, 32 not in arrest

25 of 32
non-arrest patients arrived at SpO2 94% or better
4 arrived under 90%

2 of 32
arrived with a systolic under 90
hypotension after intubation was uncommon

> **We're showing this because almost nobody does. NIV is the headline. The intubated patients are where the work still is.**

WAS IT SAFE, AND WAS IT EVERYONE?

One NIV failure in 79. And the whole staff is using it, not a few eager medics.

1 / 79

went to an intubation in the field

April 2025, after 20 minutes on the mask. CHF Exacerbation who was RSI'd successfully after BiPAP Trial and IV Nitro.

35 / 40

paramedics have used the ventilator

27 have started NIV themselves. No one medic is more than 13% of the cases.

64 min

Average first contact to hospital door

142 hours of contact time on ventilated patients, all told.

IF YOU ASK US **"Isn't NIV just delaying the tube?"**

One in 79 required an intubation in the field, none in the 18 months since. What we see at handoff is the ED RT keeping our settings, or trialing the patient off the mask on arrival. What happens after that is the follow-up we're still building.

ACROSS THE COUNTY

Ventilator patients by location

147

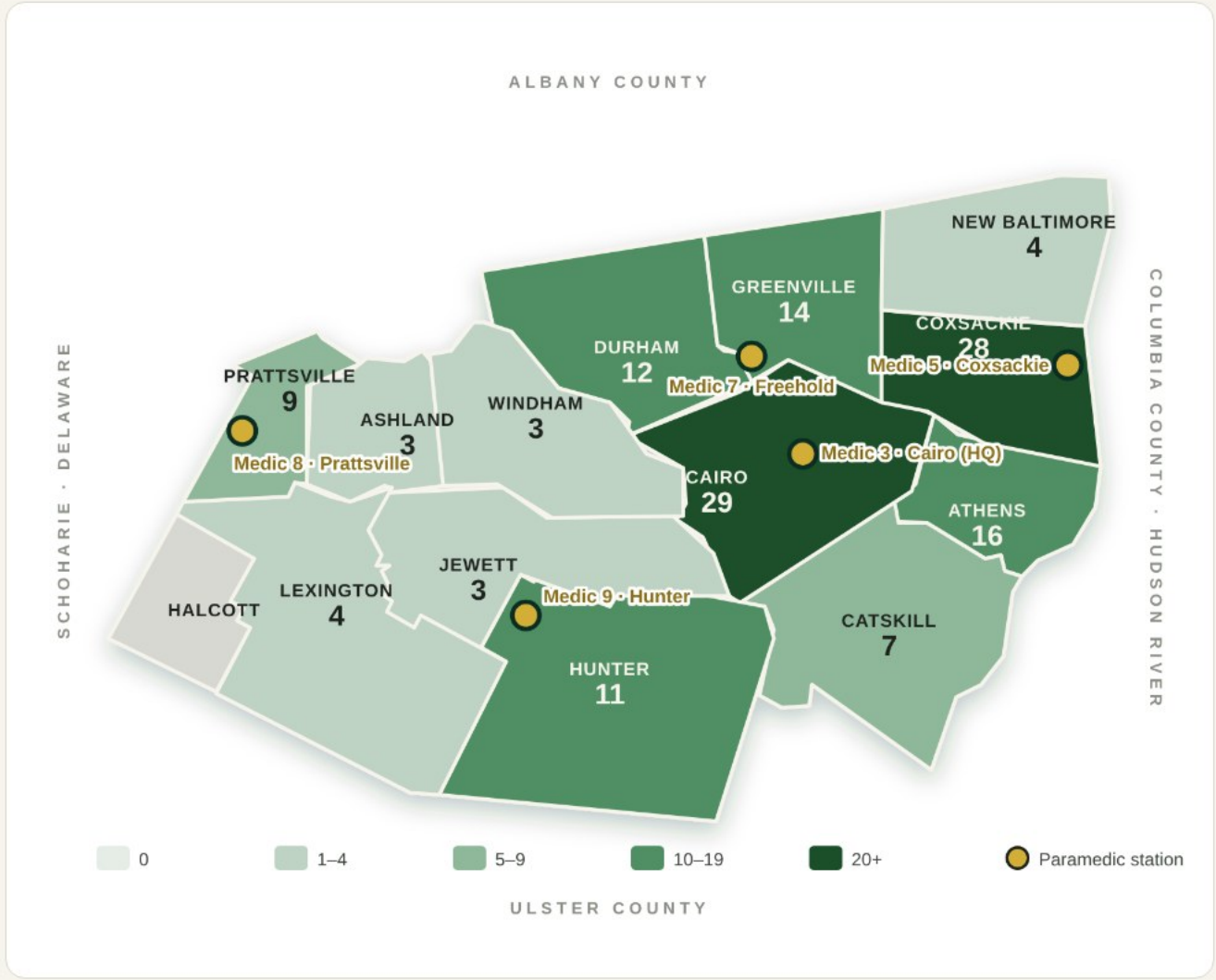
patients with a scene location; two were mutual aid into Schoharie County

13 of 14

Greene County towns with a ventilated patient

23%

on the mountaintop, furthest from the hospital



CARDIAC ARREST

Cardiac arrest survival, 2022 to 2026

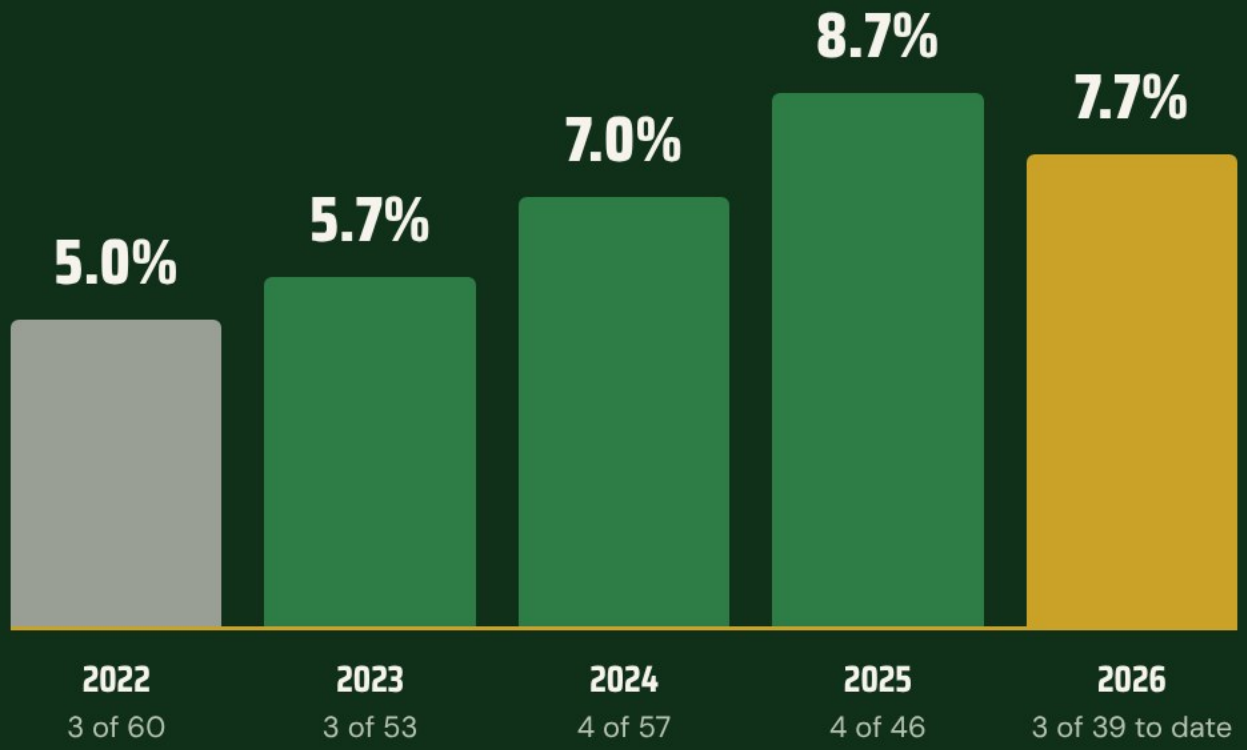
Cardiac arrest calls in Greene County have fallen by a third since 2022. The number of patients who survive to go home has not: three or four every year. As a share of arrests, that is one in twenty before the program and nearly one in eleven since.

We aren't claiming credit for the trend. What the ventilator adds is set oxygen and CO2 targets for the whole ride after ROSC, and NIV before arrest so fewer respiratory patients fall apart on the way in.

These are small numbers. One patient moves the rate about two points, so the trend matters more than any single year. Post-ROSC EtCO2 and oxygen targets are the QA focus for 2026.

Survival to hospital discharge after cardiac arrest

Survivors ÷ arrest responses, by year · outcomes via CARES



IF YOU ASK US **"Did the vent program raise your arrest survival?"**

No. The numbers are too small to say that. What we can show is that every post-ROSC patient now rides in on set targets, and respiratory patients are on NIV before they get to the point of arresting.



CASE · 2026 · INVASIVE VENTILATION

Cardiac arrest with air transport

A cyclist found down on the roadside; everyone assumed a car had hit him. Paramedics found ventricular fibrillation, defibrillated him, intubated him, and put him on the T1. LifeNet landed, took him on our settings, and flew him out.

A brewery security camera across the street later showed what happened: a spontaneous V-fib arrest mid-ride, no vehicle involved.

Off the ventilator in 24 hours. Full recovery. Back on the bike. When the tube is the right call, the vent is what makes the transport and the handoff controlled instead of improvised.



Left: in the back of the Coxsackie ambulance on this call. Right: LifeNet on scene with Medic 7 and Town of Coxsackie EMS. Patient stories are shared without identifying details.

PEDIATRICS

One pediatric patient in 147. That is the gap we know about.

1 / 145
ventilated patients under 18 in three years

70
median age of a GCEMS ventilator patient

0
pediatric NIV patients

The one was a 13-year-old in cardiac arrest, a long way from the Pediatric ER: two paramedic, a local ambulance crew, mechanical CPR, IO access, video laryngoscopy, the T1 in ASV according to patients ideal body weight. The crew's take was simple. The ventilator took the airway off one set of hands so they could run the code while monitoring the vent, and they felt they gave that patient the highest level of care they could.

WHAT WE ARE DOING ABOUT IT

The T1 covers neonatal through adult and the pediatric settings are on the reference cards, but our providers get almost no pediatric reps. Pediatric ventilation is going into the workshop rotation so the crews have done it before they need it.

IF YOU ASK US "What about kids?"

One pediatric ventilator patient in three years and no pediatric NIV. The equipment and settings are in place; the gap is experience, and pediatric scenarios are going into the workshops. Source: ESO, 145 ventilated patients.

WHAT WE WOULD TELL ANOTHER AGENCY

Five things we got right, and the one we got wrong.

1

Start small

Three donated units on the cars with the widest coverage. Expand when the data says so.

2

Train everyone

The ventilator only helps if whoever is on duty can run it, not just the people who are into it.

3

Get non-invasive capability

We bought ventilators for intubated patients. NIV is what changed practice. Buy a platform that does both.

4

Pick your number early

Ours became respiratory-failure RSIs. If you don't pick a number up front, there's nothing to report in year three.

5

Budget for training time

Two seperate training cycles, two workshops a month. Training is the real ongoing cost; consumables are about $85 per NIV patient.

6

We didn't set up hospital follow-up before go-live.

Cardiac arrest outcomes come to us through CARES, and Albany Med's EMS outreach program gives us follow-up on most of the patients we bring them. But not every patient, and not every hospital. So three years in we can't say how many NIV patients were intubated in the ED, how many went to the ICU, or what our ventilator complication rate looks like from the hospital side. That should have been in the launch plan. We're building that agreement with CMH and Albany Med now.

WHAT'S NEXT · AND WHO MADE THIS POSSIBLE

Where the program goes from here

Pediatric ventilation training

One pediatric ventilator patient in three years means our crews get almost no reps. Next is building pediatric ventilation into the workshop rotation, so if we ever need it outside a cardiac arrest the crew has done it before.

Infusion pumps on every car

Sapphire pumps for controlled sedation and pressors on a ventilated patient over a long transport.

Point-of-care labs, being explored

Handheld blood analyzers on the cars, with a draw before and after intubation and before and during NIV, so the ED has a prehospital baseline to read its own labs against.

Annual data review

The full three-year review is public at **ventilator.greene-ems.com**.

WITH THANKS TO

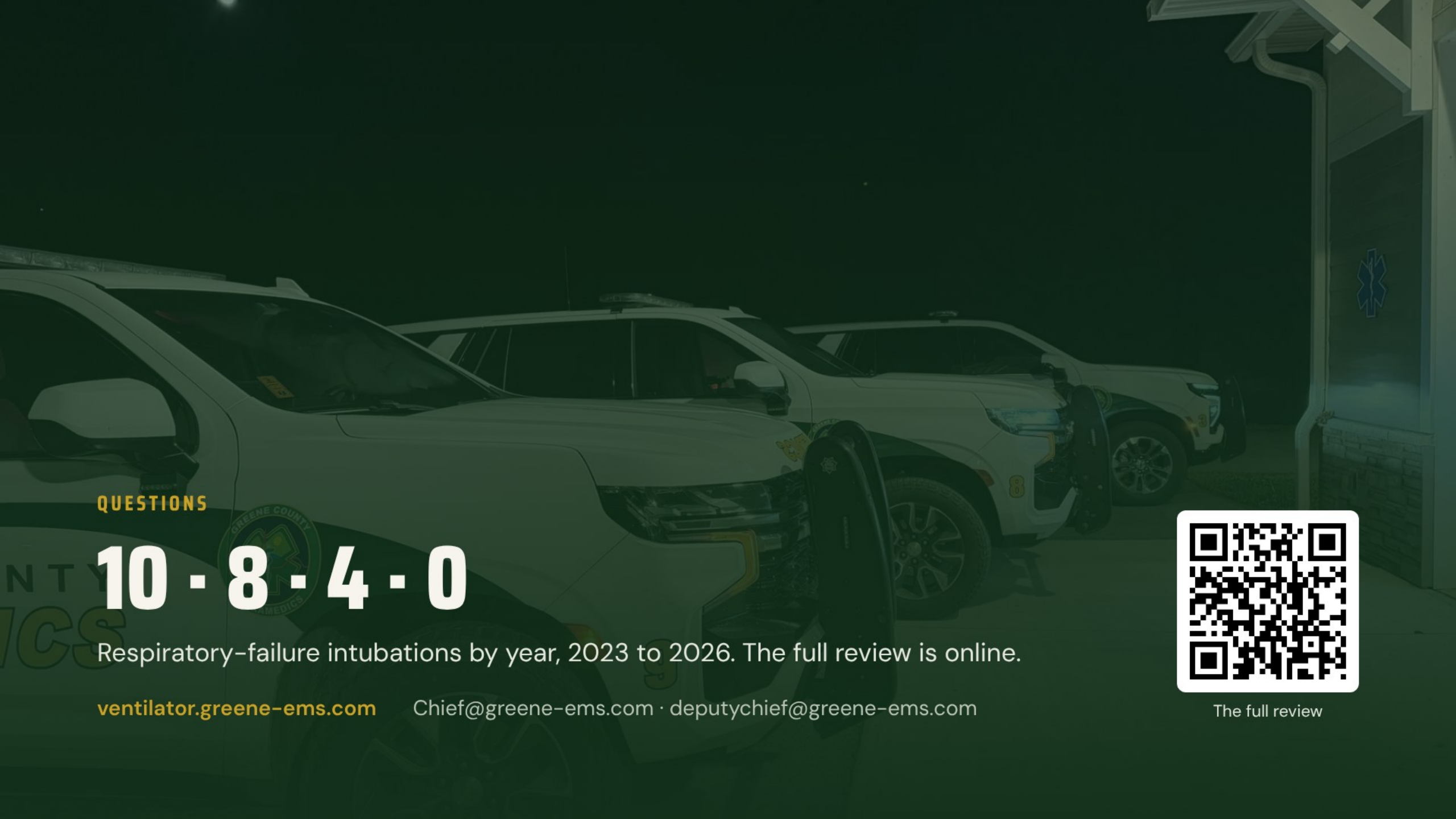
Columbia Memorial Health — the LTV 1200s that started it

Hamilton Medical — the T1, and the training to run it

LifeNet of New York — initial training and the handoffs since

ventilator.greene-ems.com
The full three-year review.

Dr. Craig Stanger, Medical Director · Mark Foster, FP-C, ALS Coordinator · the GCEMS Board of Directors · and the forty paramedics who completed every training cycle.



QUESTIONS

10 · 8 · 4 · 0

Respiratory-failure intubations by year, 2023 to 2026. The full review is online.

ventilator.greene-ems.com

Chief@greene-ems.com · deputychief@greene-ems.com

The full review